Netflix Customer Churn and Engagement Analytics

Camille : For our midterm project, we selected the Netflix Customer Churn and Engagement Analytics dataset from Kaggle, which contains 5,000 customer records and 14 columns. This dataset provides information about customer demographics, subscription plans, viewing habits, monthly fees, and overall engagement. We chose this dataset because we are interested in understanding how customer behavior influences subscription cancellations and how businesses can use this information to make better decisions. Our primary goal is to identify patterns and factors that contribute to customer churn. We plan to use Python to perform exploratory data analysis, prepare and clean the dataset, and create visualizations to better understand the relationships between different variables. As we progress toward our final project, we hope to develop and compare machine learning models that can predict whether a customer is likely to cancel their subscription. Ultimately, we want to demonstrate how machine learning can help businesses understand customer behavior, improve customer satisfaction, and develop effective retention strategies.

Min Ma Kha

The Netflix Customer Churn & Engagement Analytics dataset from Kaggle will be utilized for this project, with 5,000 records belonging to subscribers and covering 14 columns such as demographic data, payment methods, plan tiers, and watching activity. This dataset has a binary target of churned at a ratio of 50.3% to 49.6%. The aim is to predict which subscribers are going to churn instead of gauging how many of them are going to leave, since only the named list permits an action by the retention team before issue occurs. For the midterm project, the data will be split 70/30 prior to any analysis, and exploratory data analysis will only be implemented on 70% of the data set. After that, a full preprocessing pipeline will be applied: imputation, ordinal and one-hot encoding, feature engineering, log normalization, scaling and class balancing. In the final report, logistic regression, K-Nearest neighbor, decision trees, random forests, and SVM classifiers will be trained and compared.

Chris: In this project, Netflix Customer Churn and Engagement Analytics is analyzed using machine-learning methods to gain insight into the behavior of customers and predict the likelihood of a customer to churn. We use a Kaggle customer churn and engagement dataset with 5,000 rows and 14 original features. In this analysis, we investigate the relationship between customer characteristics and subscription information as well as features that describe customer engagement such as watch hours and days since last log-in. For the machine-learning task, which is a supervised binary classification, we use the feature churned as the target variable. We will compare several learning methods, i.e., Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest, and Support Vector Machine. Our evaluation criteria will be accuracy,

precision, recall, F1-measure, confusion matrix, and ROC-AUC. Furthermore, we will also address several issues with the input data, such as data quality, feature redundancy, etc. Finally, we will also discuss limitations of our results given that the input data most likely is simulated. The goal of this project is to show how input data is prepared; how exploratory data analysis is done and how machine learning methods are used to gain deeper insights into customer churning at Netflix and support data-driven customer retention decisions.